

12345678

A

Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Sensors

File: sensors.kicad_sch

GNSS (u-blox MAX-M10S) and microSD map storage

File: gnss_storage.kicad_sch

Interfaces

File: Interfaces.kicad_sch

B

C

D

E

F

12345678

F

MagNav: measure Earth's crustal magnetic anomaly with the magnetometer, subtract the core field (WMM), match against the WDMAM map on the SD card with a particle filter fused with IMU dead-reckoning -> GNSS-denied position. GNSS module gives initialisation and truth for testing.

Phone UI over WiFi AP: pick start/destination on the offline map, great-circle route + waypoints sent to the flight controller (MAVLink).

Prototype hardware – not certified for any regulated use.

ESP32-S3 + BMI088 + LIS3MDL + BMP280 + MAX-M10S + microSD

DroneMagNav Project

Sheet: /

File: dronemagnav.kicad_sch

Title: DroneMagNav – magnetic anomaly navigation board for drones

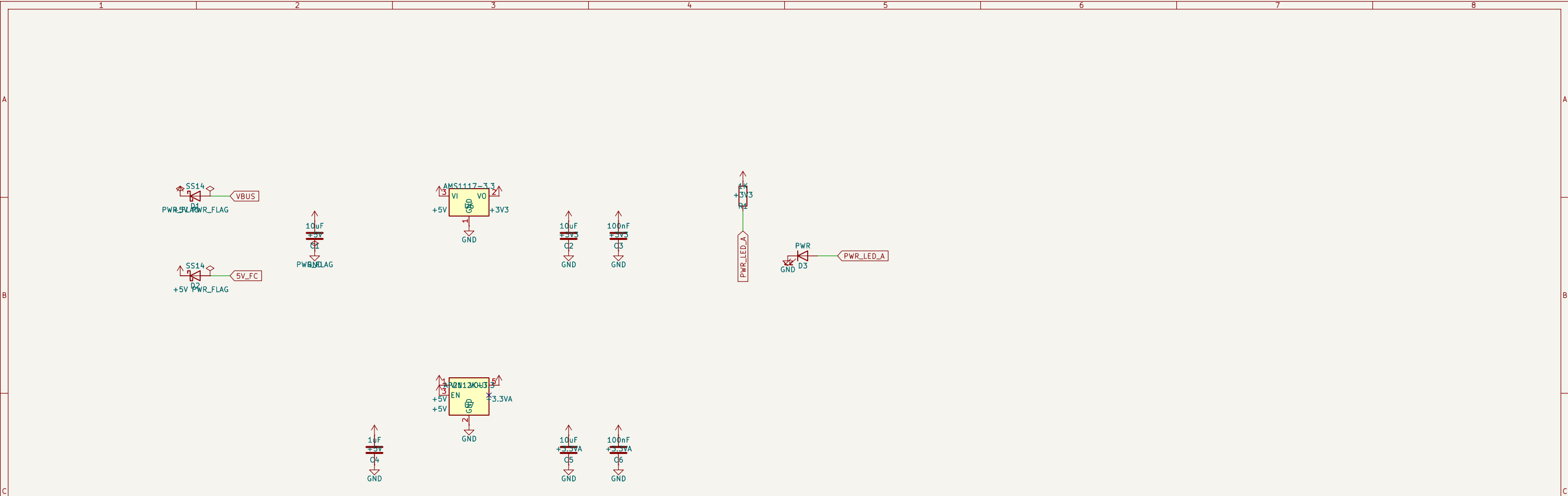
Size: A3

Date:

Rev: A

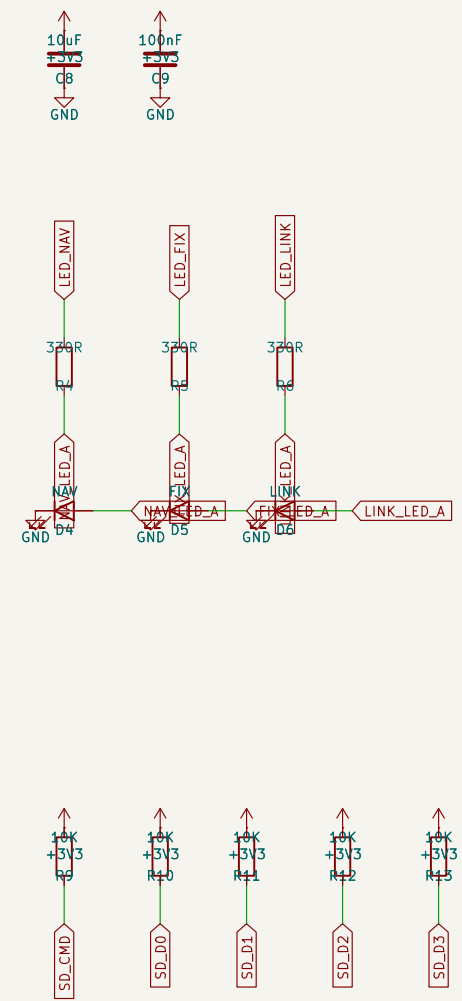
KiCad E.D.A. 10.0.3

Id: 1/6

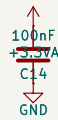
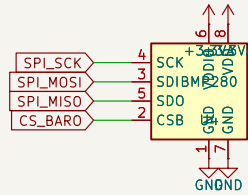
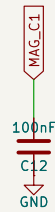
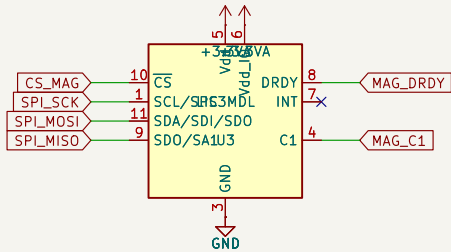
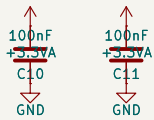
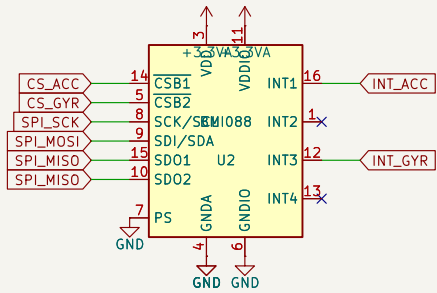


Two 5 V sources diode-ORed: USB-C VBUS (bench) and the flight-controller TELEM port (5V_FC, in flight). SS14 drop ~0.35 V.

AMS1117 (1 A) feeds ESP32/SD/GNSS; AP2112K (low-noise LDO) feeds the IMU/magnetometer/baro on the +3V3A rail.

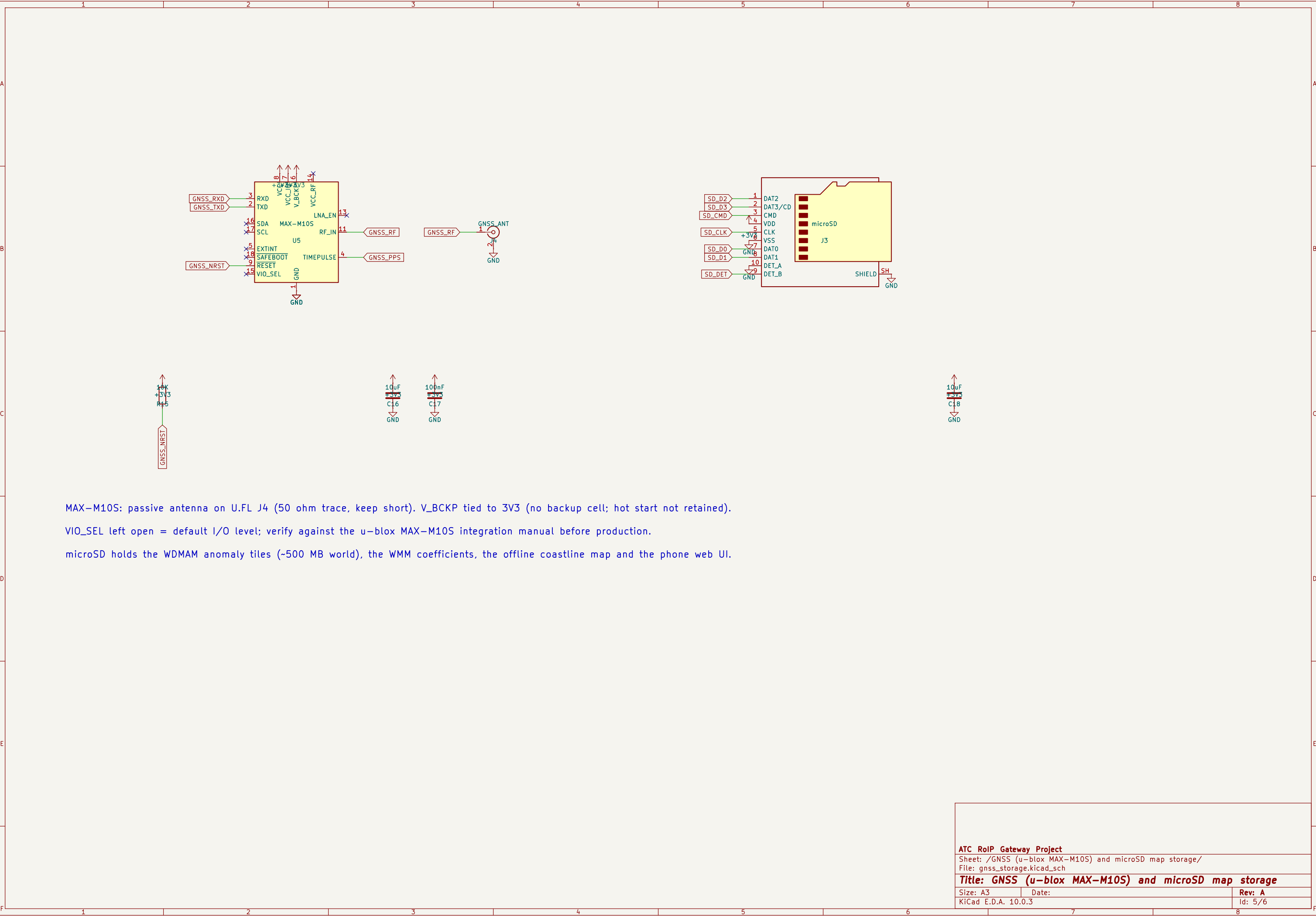


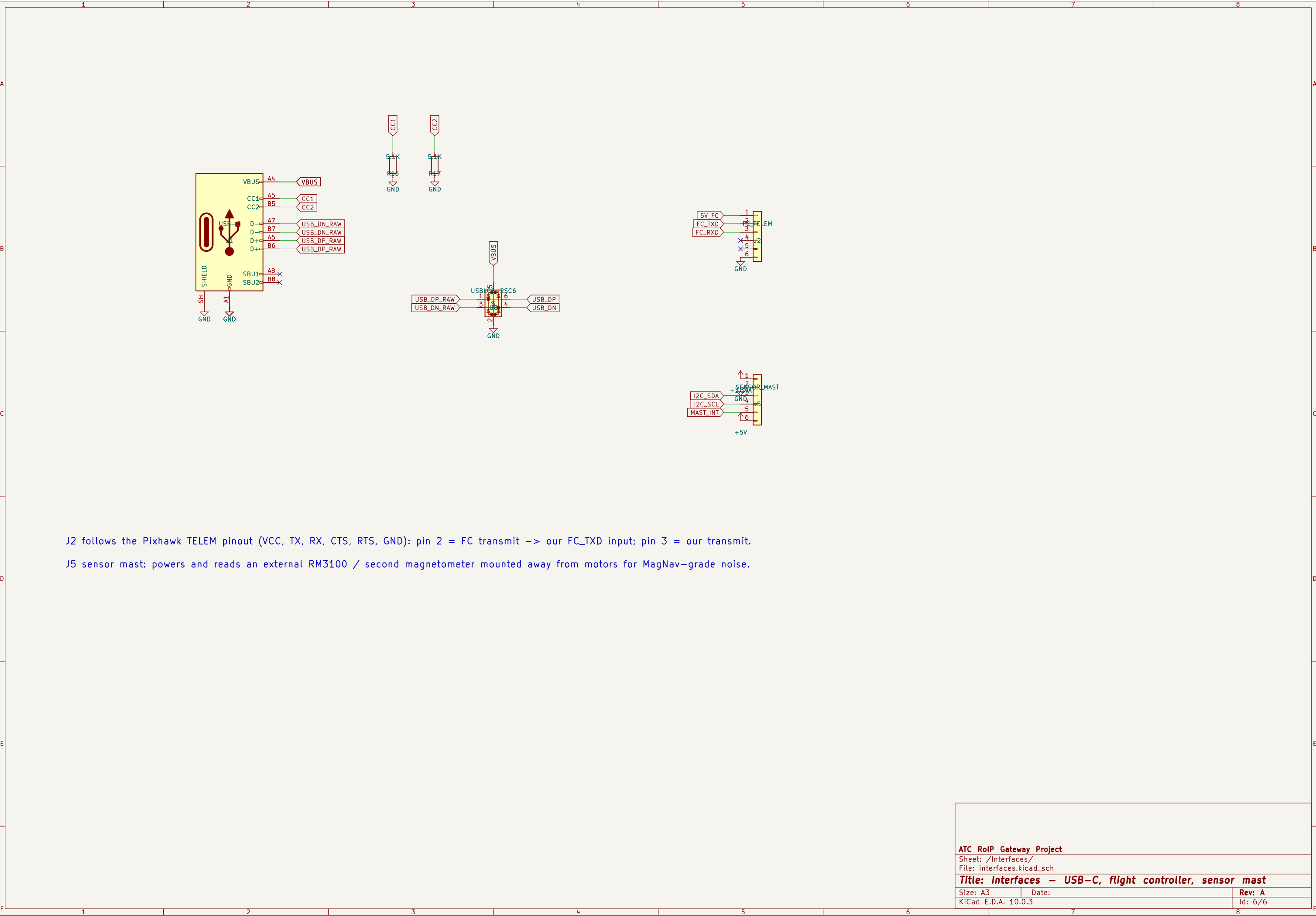
SPI @10 MHz to IMU/mag/baro; SDMMC 4-bit to microSD; UART0 pads (GPIO43/44) GNSS; UART2 MAVLink to flight controller; I2C to external sensor mast.



All three sensors share SPI (PS=GND on BMI088 selects SPI). LIS3MDL: 16-bit, ± 4 gauss, ~ 3 mG rms @80 Hz $\rightarrow \sim 40$ nT after averaging: adequate for low-altitude anomaly matching.

Keep the magnetometer ≥ 10 mm from switching parts and away from ESC/motor leads. MagNav-grade upgrade: RM3100 on the mast connector J5 (I2C).





J2 follows the Pixhawk TELEM pinout (VCC, TX, RX, CTS, RTS, GND): pin 2 = FC transmit -> our FC_TXD input; pin 3 = our transmit.

J5 sensor mast: powers and reads an external RM3100 / second magnetometer mounted away from motors for MagNav-grade noise.